

How to Get an LLM API Key and Use It with erpilot (Complete Guide)

Applies to erpilot v3.14+ Target audience: non-technical users Estimated time: 5–10 minutes

1. Why Do I Need an API Key?

erpilot's "AI Conversational CRUD" feature (query/add/edit/delete by talking) is powered by a **Large Language Model (LLM)**. LLMs run in the cloud — to call one, you must first **sign up for an account and apply for an API Key** (like a key).

What If I Don't Have an API Key?

erpilot is **designed to work without one**:

Feature	API Key Required?
Login / account management	✗ No
Inventory / Purchase / Sales / Production (mouse-click ops)	✗ No
File upload / report download	✗ No
Load demo data	✗ No
War-room real-time dashboard	✗ No
AI Assistant chat (CRUD)	✓ Yes

In other words: **You can use erpilot as a traditional ERP without applying**, but you lose the "talk to it" headline feature.

2. 3 LLM Providers Compared (Pick One)

Provider	Price	Free Tier	Chinese Quality	Recommended For
cn DeepSeek (recommended)	\$0.14 / 1M input tokens (dirt cheap)	Small credit on signup	★★★★★	Most users: best value
us OpenAI (GPT-4o-mini)	\$0.15 / 1M	\$5 trial (90 days)	★★★★	Already have OpenAI account
us Anthropic Claude	\$0.25 / 1M (haiku)	\$5 on signup	★★★★★	Better reasoning, long-form
🏠 Ollama (offline)	\$0 (runs locally)	Unlimited	★★★	Have GPU + want fully offline

Our Recommendation → DeepSeek

Reasons:

1. 100× cheaper than OpenAI
2. Best Chinese understanding (especially Traditional Chinese)
3. OpenAI-compatible API format — easy to switch providers later
4. A typical small factory's monthly bill is NT\$10–100 (essentially free)

If you're worried about a Chinese company storing your data → pick OpenAI or Claude (data stays in the US).

3. Apply for a DeepSeek API Key (Recommended, 5 min)

Step 1 · Sign Up

1. Open https://platform.deepseek.com/sign_up
2. Register with **email** or **phone number** (Gmail works best)
3. Verify email, set password, land on home page

Step 2 · Get API Key

1. After login, click "API Keys" in left sidebar
2. Click blue "Create new API Key" button (top right)
3. Name the key (e.g., `erpilot-2026`), click Create

4. Copy the `sk-xxxxxxxxxxxxxxxxxxxxxxxxxxxx` shown immediately (shown only once!)

⚠️ Forgot to copy → delete and create a new one. Don't try to find it in the page.

Step 3 · Top Up (Optional)

The free signup credit is limited (about NT\$15 / ~200 chats).

To continue using:

1. Click "Top Up" in left sidebar
2. Minimum top-up is USD\$2 (~300 chats)
3. Credit card / PayPal / Alipay / WeChat Pay all accepted

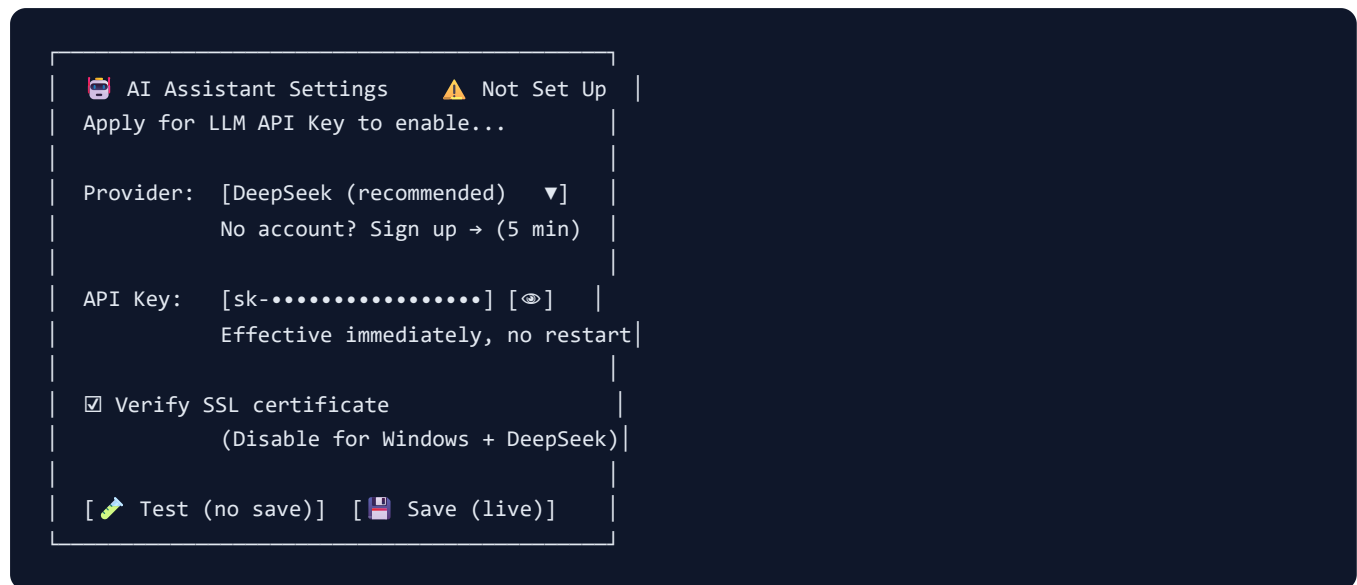
💡 **Saving tip:** NT\$30/month is plenty for a small factory's ERP chat volume. Try the free tier first to see how quickly you burn through it.

4. Give the Key to erpilot (3 Methods)

Method A: Settings Page (recommended, easiest)

1. Login to erpilot (<http://localhost:5173>)
2. Click "⚙️ Settings" in left sidebar
3. In the top "🤖 AI Assistant Settings" section:
 - **Provider** → pick "DeepSeek" from dropdown
 - **API Key** field → paste your `sk-xxxxx...`
 - Click "🔧 Test Connection (no save)" → see "✅ DeepSeek connection successful"
 - Click "💾 Save (effective immediately)"
4. Done! No restart, no re-login. Go to AI Assistant page and start talking.

Screenshot Mockup (ASCII art)



Method B: Edit backend/.env (for developers)



Save, then:

- Docker mode: `docker compose restart backend`
- Dev mode: run `stop_dev.bat` → `start_dev.bat`

Method C: Environment Variables (advanced / production)



5. Windows + DeepSeek Common Issue: SSL Certificate Error

Symptom

Click "Test Connection" and see:

 Connection failed: SSLCertVerificationError

Cause

Windows' CA cert store hasn't updated with DeepSeek's root certificate. Common issue with Windows + DeepSeek combo — not an erpilot bug.

Fix (fastest)

In erpilot Settings, uncheck "Verify SSL certificate" → retry test. This just makes Python skip SSL verification — doesn't affect your security (key is still sent over TLS-encrypted connection).

Cleaner fix: upgrade Python `certifi` package — not necessary for non-developers.

6. Other Provider Steps

OpenAI (GPT-4o-mini)

1. Sign up: <https://platform.openai.com/signup> (credit card required)
2. Get Key: left API Keys → Create new secret key (shown once)
3. In erpilot Settings: pick OpenAI, paste key, save

Anthropic Claude

1. Sign up: <https://console.anthropic.com/>
2. Go to Workbench → Settings → API Keys → Create Key
3. In erpilot Settings: pick Anthropic, paste key, save

Ollama (local offline)

1. Download: <https://ollama.com/download> → install
2. Terminal: `ollama pull llama3.2` (~2GB model)
3. In erpilot Settings: pick Ollama, leave key empty, save
4. Confirm `ollama serve` is running locally

7. Security Reminders

✓ Things You Should Do

- API Key = password; leaks let others use your quota
- Don't paste keys into chat / Slack / GitHub commits
- Rotate keys regularly (every 3-6 months)
- If you suspect a leak → revoke at provider + create new

✓ How erpilot Handles Your Key

- Stored in `backend/.env` on your local machine (plaintext file)
- **Never** uploaded to any server
- **Never** git-committed (`.gitignore` excludes `.env`)
- Pre-commit hook auto-scans `sk-` patterns and blocks accidents

✓ Where Conversation Data Goes

When you chat with the AI Assistant:

- Your **typed text** is sent to the LLM provider (DeepSeek/OpenAI/Anthropic)
- **Business DB data is not sent** (unless the AI calls a tool that queries DB and includes results in subsequent context)
- Provider retention policies differ — check their privacy terms

To keep data fully on-premise → use **Ollama offline mode** (Method D).

8. Cost Estimates (Small Factory Reality)

Scenario	Usage	DeepSeek/mo	OpenAI/mo
Boss asks 5× daily	150/month	NT\$3	NT\$45
Sales rep uses 10× daily	1500/month	NT\$30	NT\$450
Whole 20-person company heavy use	20000/month	NT\$400	NT\$6,000

💡 **erpilot is token-efficient:** each chat usually < 2000 tokens.

9. Troubleshooting

Symptom	Fix
"Connection refused"	Check network / proxy settings
"Invalid API key"	Re-confirm key has no missing chars / extra spaces
"Rate limit"	Too many requests, wait minutes or upgrade plan
"Insufficient balance"	DeepSeek balance empty, go Top Up
"Model not found"	Provider changed model name; update in Settings
Windows SSL error	Uncheck "Verify SSL certificate"
Settings won't save	Confirm you're admin (need <code>system.config.update</code>)
Key change has no effect	Save is instant; if not, hard refresh (Ctrl+F5)

10. Related Documentation

- [USER_MANUAL_EN.md](#) — Full user manual
- [PRODUCT_OVERVIEW_EN.md](#) — Product overview for buyers
- [INSTALLATION_EN.md](#) — Installation guide
- [LICENSE-SMALL-BUSINESS.md](#) — Free tier (≤20 users)

Chinese version: [HOW_TO_GET_LLM_API_KEY_ZH.md](#)